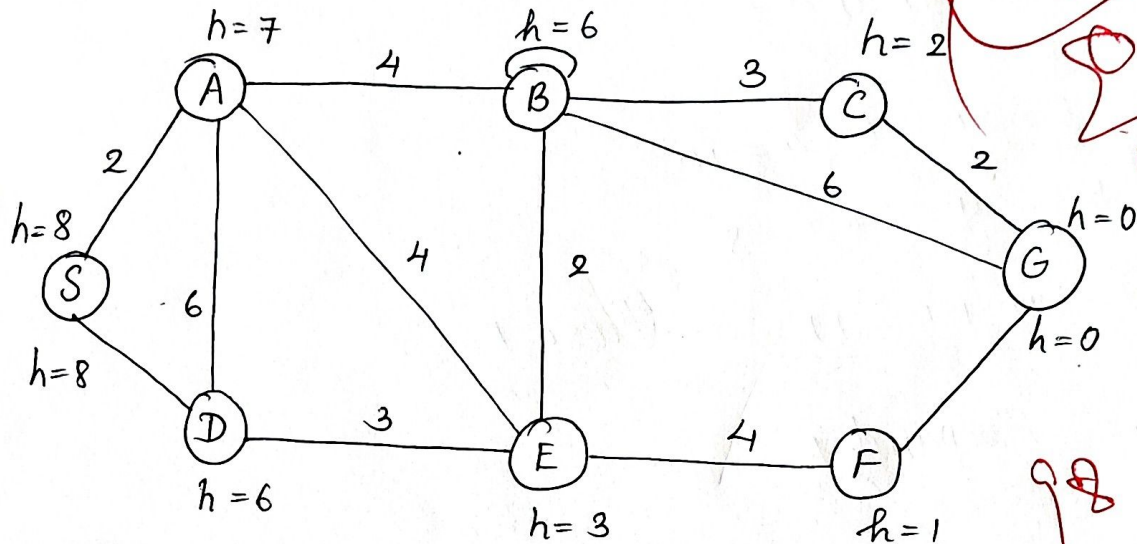


1) Smart Ambulance Navigation:



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 98

Algorithm: A* Search

function:

$$f(n) = g(n) + h(n)$$

where,

$g(n)$ = actual cost from S to n

$h(n)$ = heuristic estimate from n to G.

Step 1:

$$g(s) = 0 \quad h(s) = 8$$

$$f(s) = 0 + 8 = 8$$

$$S \rightarrow A \Rightarrow g(n) = 2 \quad h(n) = 7 \Rightarrow f(n) = 9$$

$$S \rightarrow D \Rightarrow g(n) = 6 \quad h(n) = 6 \Rightarrow f(n) = 12$$

$f = 9$ choose A

Step 2: Expand A

$$A \rightarrow B: g = 2 + 4 = 6$$

$$A \rightarrow D: g = 2 + 6 = 8$$

$$A \rightarrow E: g = 2 + 4 = \underline{6} \quad \text{choose E}$$

Step 3: Expand E

$$E \rightarrow F: g = 6 + 4 = \underline{10}$$

$$E \rightarrow B: g = 6 + 2 = 8$$

$$f(F) = 10 + 1 = 11 \quad \text{choose F}$$

Step 4: Expand F

$$F \rightarrow G \Rightarrow 2$$

$$g(G) = 10 + 2 = 12$$

$$f(G) = 12 + 0 = 12$$

Step 5:

B's route:

$$g(B) = 6, f(B) = 6 + 6 = 12$$

$$B \rightarrow C: g(C) = 6 + 3 = 9$$

$$f(C) = 9 + 2 = 11$$

Step 6: Expand C

$$C \rightarrow G: g(G) = 9 + 2 = 11$$

$\therefore$ The better ^{optimal} route is

$$S \rightarrow A \rightarrow B \rightarrow C \rightarrow G$$

$$\text{Total cost: } 2 + 4 + 3 + 2 = 11$$

It is Optimal route.

Alternative but not optimal route

$$S \rightarrow A \rightarrow E \rightarrow F \rightarrow G$$

$$\text{cost: } 2 + 4 + 4 + 2 = 12$$

It is not optimal

2) Autonomous Warehouse Robot:

5	S	?	?	?	?	?	?
4	?	#	?	?	#	?	?
3	?	?	?	#	?	?	#
2	?	#	?	?	?	#	?
1	?	?	#	?	?	?	G
	1	2	3	4	5	6	7

S charging station (start)

G Target shelf (Goal)

Obstacle (Unknown initially)

? Unknown cell

Strategy: ^{Online} Greedy Best-First Search using Manhattan distance

$$h(n) = |x_n - x_G| + |y_n - y_G|$$

Goal :

$$G = (7, 1)$$

Step	Robot position	Decision / Observation
1)	S(1,5)	Move right $\rightarrow$ (2,5)
2	(2,5)	Move right $\rightarrow$ (3,5)
3	(3,5)	Move right $\rightarrow$ (4,5)
4	(4,5)	Move right $\rightarrow$ (5,5)
5	(5,5)	Move right $\rightarrow$ (6,5)
6	(6,5)	Move right $\rightarrow$ (7,5)
7	(7,5)	Move down down $\rightarrow$ (7,4)
8	(7,4)	Discovers obstacle at (7,3) cannot move down.

9	(7, 4)	Replan: move left $\rightarrow$ (6, 4)
10	(6, 4)	Move down $\rightarrow$ (6, 3)
11	(6, 3)	Discovers obstacle at (6, 2) cannot move down
12	(6, 3)	Replan: move left $\rightarrow$ (5, 3)
13	(5, 3)	Move down $\rightarrow$ (5, 2)
14	(5, 2)	Move down $\rightarrow$ (5, 1)
15	(5, 1)	Move right $\rightarrow$ (6, 1)
16	(6, 1)	Move right $\rightarrow$ G(7, 1)

Final path: S $\rightarrow$ (2, 5) $\rightarrow$ (3, 5) $\rightarrow$ (4, 5) $\rightarrow$ (5, 5)
 $\rightarrow$ (6, 5) $\rightarrow$ (7, 5) $\rightarrow$ (7, 4) $\rightarrow$ (6, 4)
 $\rightarrow$ (6, 3) $\rightarrow$ (5, 3) $\rightarrow$ (5, 2)

Total moves = 14

When an obstacle is discovered

Detect obstacle $\rightarrow$ Update map $\rightarrow$

Recalculate $h(n)$ $\rightarrow$ choose new move

Online search vs Offline search: Online search makes decisions using currently known information and replans when new obstacles appear. Offline search assumes the complete map is known beforehand.

3) University Examination Scheduling.

a) Variables:

$E1, E2, E3, E4, E5, E6$

Domain of each variable:

$$D(E_i) = \{T1, T2, T3, T4\}$$

Student Conflict constraints:

$$E1 \neq E2, E1 \neq E3, E2 \neq E4, E2 \neq E5, \\ E3 \neq E5, E4 \neq E6$$

Hall constraints:

Exam	Students	Possible halls
E1	100	A
E2	80	A, B
E3	70	A, B
E4	60	A, B
E5	90	A, B, C
E6	50	A, B
		A, B, C

b) MRV and forward checking

Step 1:

$$E2 = T2$$

Variable

Remaining domain

E1

T1, T3, T4

E2

T2

E3

T1, T2, T3, T4

E4

T1, T3, T4

E5

T1, T3, T4

E6

T1, T2, T3, T4

Step 2:

$$E1 = T1$$

Variable

Remaining domain

E1

T1

E2

T2

E3

T2, T3, T4

E4

T1, T3, T4

E5

T1, T3, T4

E6

T1, T2, T3, T4

Step 3:

$$E3 = T2$$

Variable

Remaining domain

E1

T1

E2

T2

E3

T2

E4

T1, T3, T4

E5

T1, T3, T4

E6

T1, T2, T3, T4

Step 4:

$$E4 = T1$$

~~Variable~~

Remaining domain

~~E1~~

T1

~~E2~~

T2

~~E3~~

T2

~~E4~~

T1

~~E5~~

T1, T3, T4

~~E6~~

T2, T3, T4

Step 5:

$$E5 = T3$$

Variable

Remaining domain

E1

T1

E2

T2

E3

T2

E4

T1

E5

T3

E6

T2, T3, T4

Step 6:

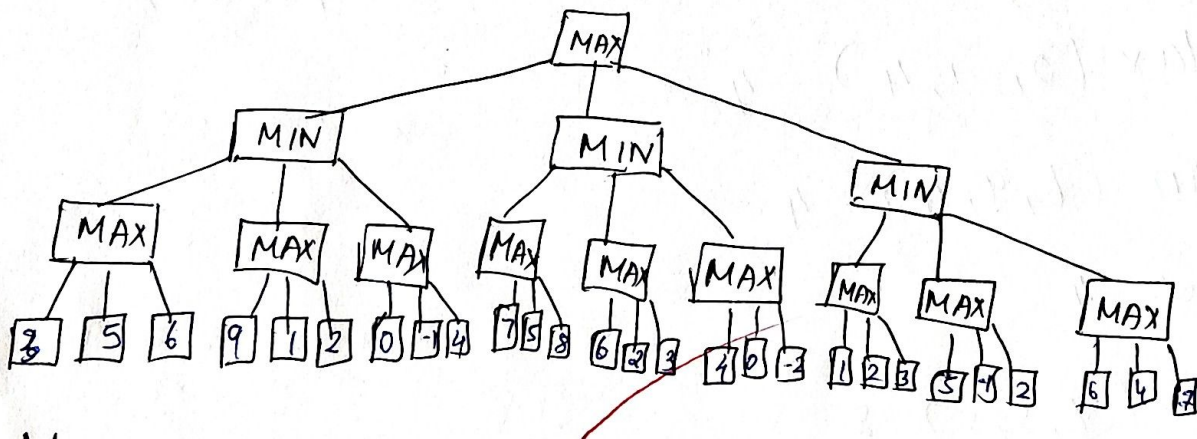
$$E6 = T2$$

E6 conflicts ^{only} with E4, which is at T1.

c) Final Timetable

Time slot	Examination	Hall	Students
T1	E1	Hall A	100
T1	E4	Hall C	60
T2	E2	Hall B	80
T2	E3	Hall A	70
T2	E6	Hall C	50
T3	E5	Hall B	90
T4	—		

4) Game - Playing AI:



Minimax:

$$\text{Min}(6, 9, 4) = 4$$

$$\text{Min}(8, 6, 4) = 4$$

$$\text{Min}(3, 5, 7) = 3$$

Root Max decision

$$\text{MAX}(4, 4, 3) = 4$$

AI can choose either the left or middle move, both getting minimax value of 4.

Alpha-Beta Pruning

Evaluate nodes left to right

$$\text{Max}(3, 5, 6) = 6$$

$$\beta = 6$$

$$\text{Max}(9, 1, 2)$$

$$\alpha = 9, \beta = 6$$

$$\alpha \geq \beta$$

remaining values 1 and 2 are pruned

$$\text{Max}(0, -1, 4) = 4$$

$$\text{Min}(6, 9, 4) = 4$$

$$\alpha = 4$$

$$\text{Max}(7, 5, 8) = 8$$

$$\text{Max}(6, 2, 3) = 6$$

$$\text{Max}(4, 0, -2) = 4$$

$$\text{Min}(8, 6, 4) = 4$$

$$\alpha = 4$$

$$\text{Max}(1, 2, 3) = 3$$

$$\beta = 3$$

$$\beta \leq \alpha$$

Branches pruned (5, -1, 2) and (6, 4, 7)

Minimax value = 4

Best move = left or middle

(10)

Alpha-Beta pruned

- Values 1, 2 from second max node of left subtree.
- Entire branches (5, -1, 2) and (6, 4, 7) from right subtree.

Delivery Route Optimization

Distance matrix (in km)

	D	A	B	C	E
D	-	6	9	8	7
A	6	-	3	5	6
B	9	3	-	4	5
C	8	5	4	-	3
E	7	6	5	3	-

Strategy: Hill climbing

Objective: Minimize total route distance

Step 1: Initial solution

Given:

S5 : $D \rightarrow A \rightarrow E \rightarrow B \rightarrow C \rightarrow D$

Cost : 23 km

Step 2:

Best neighbour. cost = 22 km

$22 < 23$

S5 $\rightarrow$ neighbour

Step 3:

All neighbouring solutions have cost ≥ 22 km.

Therefore, no further improvement is possible

Final route = best neighbour of S5

Final cost = 22 km

The route is a local optimum because no immediate neighbouring route has a lower cost

Can it be a global optimum?

Hill climbing checks only neighbouring solutions.

A better route may exist outside the current neighbourhood.